Acknowledgements

....

Contents

Chapter 1

Introduction

1.1 Motivation

Financial time series data is never fully observed. Limit-order book feeds drop quotes during bursts of order cancellations. Equity markets suspend trading when prices move too violently. Over-the-counter bond markets leave entire yield-curve tenors unquoted for days. Implied volatility surfaces contain large regions where no liquid option trade has occurred. Macro and alternative data arrive at mixed, irregular frequencies, with publication lags that create structural gaps at the daily level. This missing data is pervasive and often elusive, and gets in the way of advanced analysis. Thus practitioners or a researchers face the same decision: what value should be attributed to the missing observation, and how much uncertainty should be attached to that attribution?

The simplest and most common solution is to carry forward the last observed value, but this can lead to systematic mispricing, underestimation of risk, and distortion of correlations among assets.

We aim to take a different path, one that looks for solutions that aim to preserve as much as possible the statistical properties of the series, in order to maintain the integrity of the underlying data and consequently the robustness of any downstream model that relies on it. Since this can be considered an inference problem, we will also explore ML algorithms, and in classical machine learning approach, we will focus on quality of inference rather than interpretability of the model.

In this thesis we will test these different methods of imputation, and evaluate them not only on the accuracy of the imputation itself (MSE,...), but also on the quality of the downstream model that uses the imputed data, and how far it deviates from the case where all data is observed.

1.2 Literature review

The literature on missing data is vast and spread across statistics, econometrics and, more recently, machine learning. Rather than attempting an exhaustive survey, we trace the ideas this thesis directly builds on: the foundational framework that defines what missingness is, the classical likelihood-based methods with the EM algorithm at their core, the progressive shift towards algorithmic and deep learning imputers, and finally how these tools have been absorbed by economics and finance.

1.2.1 Foundations

Dropping incomplete records, or filling them with some convenient value and proceeding as if nothing had happened, has always been common practice. It was not, however, the only option available, and the problem was attacked seriously long before it acquired its modern formulation. In the design-of-experiments tradition of the 1930s an incomplete field trial was already handled by an explicit estimate of what had not been observed. Allan and Wishart (1930) gave the first missing plot formula: for a randomized block design they derived the value which, once inserted in the empty cell, leaves the treatment comparisons of the analysis of variance unbiased. Yates (1933) extended the device to any number of missing plots and to more general layouts, and identified the principle behind it, namely that the substituted values are exactly those that minimize the residual sum of squares. He also observed that the degrees of freedom must be reduced by the number of values supplied, since a filled cell carries no information of its own; the same warning applies to every single imputation in this thesis. Bartlett (1937) then recast the computation as an analysis of covariance with one dummy covariate per missing observation, which returns the estimate and the corrected residual sum of squares in a single pass.

Outside designed experiments the work started just as early. Wilks (1932) studied the moments of estimates computed from fragmentary samples of a multivariate normal population, Anderson (1957) later obtained closed-form maximum likelihood estimates when the pattern of missingness is nested, and Buck (1960) proposed filling each gap with a regression prediction from the variables observed on the same unit, a method he wrote for use on an electronic computer. Computers also turned the iterative version of Yates's method into a general algorithm: Healy and Westmacott (1956) start from a guess for the missing entries, fit the model by least squares, replace the entries by their fitted values, and iterate until the values stabilize. Dempster et al. (1977) would later list this scheme among the special cases their algorithm unifies. By the time Afifi and Elashoff (1966) reviewed the field there was already a substantial body of techniques, and Orchard and Woodbury (1972) had extracted from it a missing information principle of some generality. What this work lacked was a language for the missingness itself. Every procedure carried an implicit assumption about why the entries were absent, the assumption was rarely stated, and there was no way to say when a procedure was justified and when it was not.

The turning point is Rubin (1976), who supplied that language by treating missingness itself as a random object. Given a data matrix and an indicator matrix flagging which entries are observed, the joint distribution factorizes into the law of the data and the law of the *missingness mechanism*, the conditional distribution of the indicators given the data. Rubin's classification of this mechanism is still the standard language of the field: data are missing completely at random (MCAR) when the mechanism is independent of the data; missing at random (MAR) when it depends only on the observed part; and missing not at random (MNAR) otherwise. The central result of the paper is *ignorability*: under MAR, plus a parameter-distinctness condition, likelihood-based and Bayesian inference can ignore the mechanism altogether and work with the observed-data likelihood. This single result justifies most of the methods in this thesis; whenever it fails, as in the stressed missingness pattern of Section 2.1.1, every method that relies on it stands on shaky ground.

The monograph of Little and Rubin (2019), first published in 1987, organized the field around this framework and remains its canonical reference. Their taxonomy of approaches (complete-case analysis, weighting, single imputation, likelihood-based methods) is still the way the subject is taught, and their analysis of the simple methods is directly relevant here: listwise deletion is unbiased only under MCAR and can discard a dramatic fraction of the sample, while unconditional mean imputation distorts variances and correlations even under MCAR. These are precisely the two baselines we adopt in Chapter 2, and the literature predicts their failure modes quite exactly.

Two other foundational strands deserve mention. The first is weighting: instead of imputing, one reweights the complete cases by the inverse of their probability of being observed. Robins et al. (1994) developed this idea into a full semiparametric theory, including doubly robust estimators; Seaman and White (2013) give an accessible review. The second is *multiple imputation*, introduced by Rubin (1987): draw several imputed datasets from the posterior predictive distribution of the missing values, analyse each one as if it were complete, and combine the results with simple rules that propagate the imputation uncertainty into the final standard errors (see also Rubin, 1996). Multiple imputation is the historical answer to a criticism that applies to every single-imputation method in this thesis: an imputed dataset treated as real understates uncertainty, because imputed values are estimates, not observations.

The MNAR case is fundamentally harder, since the mechanism is not identifiable from the observed data alone and any analysis rests on untestable assumptions. In econometrics the problem was attacked early and largely independently by Heckman (1979), whose sample-selection model corrects for non-randomly missing outcomes through an explicit model of the selection process; it remains the standard MNAR tool in applied economics. In biostatistics the accepted practice is sensitivity analysis across plausible mechanisms (Molenberghs and Kenward, 2007). Finally, the translation of MAR to dependent, dynamically evolving data is more subtle than it appears; Commenges and Jacqmin-Gadda (2022) give a rigorous stochastic-process formulation, which is the natural setting for financial time series.

1.2.2 Likelihood methods and the EM algorithm

Once ignorability is granted, inference reduces to maximizing the observed-data likelihood, which almost never has a closed form because the missing entries must be integrated out. The Expectation–Maximization algorithm of Dempster et al. (1977) is the fundamental tool for this task, and arguably the most important paper in the field after Rubin (1976). Its logic mirrors the structure of the problem: the E-step computes the conditional expectation of the complete-data log-likelihood given the observed data and the current parameters, and the M-step maximizes it. The paper proved that each iteration increases the observed likelihood, unified a large number of pre-existing special-purpose schemes, and named the method. The convergence theory was completed by Wu (1983), who fixed a flaw in the original proof and gave the regularity conditions under which EM converges to a stationary point. Since EM never forms the observed information matrix, standard errors require extra work; Louis (1982) showed how to recover them from complete-data quantities. Among the many refinements that followed, the ECM and ECME algorithms (Liu and Rubin, 1994; Meng and Rubin, 1993) replace the M-step with cheaper conditional maximizations; McLachlan and Krishnan (2008) is the standard book-length treatment.

For our purposes the relevant instance is EM for the mean vector and covariance matrix of an incomplete multivariate normal sample (Little and Rubin, 2019, Ch. 11): the E-step reduces to linear regressions of the missing coordinates on the observed ones, and the fixed point delivers both the maximum likelihood estimate of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and, as a by-product, conditional-expectation imputations. Since daily returns are decidedly non-Gaussian (Cont, 2001), the robust variant matters: Little (1988) derived EM for the multivariate Student-t, whose E-step simply downweights outlying observations, and Liu and Rubin (1995) made its estimation practical via ECME. This is the version we implement in Chapter 2.

The time series counterpart of this machinery is the state-space framework. The Kalman filter (Kalman, 1960) accommodates missing observations with no modification at all: when a measurement is absent, the update step is skipped and the prediction carries through. Jones (1980) exploited this to fit ARMA models to incomplete series by exact maximum likelihood, and Shumway and Stoffer (1982) closed the loop with EM: the Kalman smoother computes exactly the conditional expectations the E-step requires, so parameter estimation and imputation come out of the same iteration. Harvey and Pierse (1984) applied these ideas to economic time series, Kohn and Ansley (1986) settled exact interpolation for ARIMA models, and the monographs of Harvey (1989) and Durbin and Koopman (2012) remain the references for the general theory.

1.2.3 From multiple imputation to machine learning

On the applied side, the 1990s and 2000s standardized imputation practice. Schafer (1997) provided workable algorithms for multiple imputation under joint multivariate models, and the chained equations approach of van Buuren and Groothuis-Oudshoorn (2011) replaced the joint model with a sequence of univariate conditional models, one per variable, cycled until convergence; its flexibility made it the de facto standard in biostatistics and the social sciences. van Buuren (2018) gives a modern account of the whole practice.

In parallel, a more algorithmic tradition developed methods that dispense with an explicit probability model. k-nearest-neighbours imputation was popularized by Troyanskaya et al. (2001) for gene expression data and transfers directly to return panels: a missing entry is filled with a weighted average of the corresponding entries of similar rows (dates, in our case); random forests followed (Stekhoven and Bühlmann, 2012). A distinct and elegant line is low-rank matrix completion (Candès and Recht, 2009; Mazumder et al., 2010), which recovers missing entries under the assumption that the data matrix is approximately low-rank, a natural assumption for equity returns where a few factors explain most of the covariance.

Deep learning entered the field once recurrent networks were adapted to irregular inputs. GRU-D (Zhengping Che and Liu, 2016) feeds the missingness mask and the time since the last observation into a gated recurrent unit with learned decay; BRITS (Cao et al., 2018) treats imputations as variables of a bidirectional recurrent graph, trained by consistency between the two directions, and requires no assumption on the generating process. Generative approaches recast imputation as conditional sampling: GAIN adapts adversarial training (Yoon et al., 2018), GP-VAE combines a variational autoencoder with a Gaussian process prior in latent space (Fortuin et al., 2020), SAITS relies on self-attention (Du et al., 2023), and CSDI (Tashiro et al., 2021) uses conditional score-based diffusion, producing a full predictive distribution for the missing values rather than a point estimate, which is why we select it as our generative representative. Dong et al. (2024) survey this rapidly growing area, and Kreiss et al. (2024) bridge it with the signal processing literature.

Two caveats recur across this literature. First, essentially all deep imputers assume MAR, usually implicitly through their masked training objective; identification under more general mechanisms has been studied only very recently (Ma et al., 2025). Second, evaluation is almost universally pointwise, RMSE or MAE on artificially masked entries, which says little about whether the joint distribution, and in particular the covariance structure, survives imputation. Both caveats are starting points for this thesis.

1.2.4 Applications in economics and finance

In economics, missing data rarely goes by that name. The oldest example is non-synchronous trading: Scholes and Williams (1977) showed that stale prices, which are in effect missing prices carried forward, bias beta estimates, and derived corrected estimators. This is exactly the last-observation-carried-forward distortion described in Section 1.1.

The macroeconometric literature turned missing data handling into routine practice, because macro datasets are incomplete by construction: series are released at mixed frequencies and with staggered publication lags (the so-called ragged edge). Stock and Watson (2002) used an EM–principal-components iteration to extract factors from large unbalanced panels; Mariano and Murasawa (2003) treated quarterly GDP as a monthly series with two out of three values missing, estimated in state-space form; Giannone et al. (2008) formalized nowcasting, where the Kalman filter absorbs data releases as they arrive; and Doz et al. (2012) and Bańbura and Modugno (2014) developed maximum likelihood estimation of large factor models under arbitrary missingness patterns, essentially scaling Shumway and Stoffer (1982) to hundreds of series. This machinery runs in production at central banks today. For panels with a time dimension, Honaker and King (2010) adapted multiple imputation to time-series cross-section data, noting that off-the-shelf procedures ignore temporal smoothness.

In finance proper, the seminal reference is Stambaugh (1997), who considered assets whose return histories start at different dates, the typical shape of any real multi-asset panel, and derived maximum likelihood estimators of means and covariances that use the long histories to sharpen the moments of the short ones. The message, that discarding incomplete history throws away information about the covariance matrix, is a direct ancestor of our evaluation criteria. Kofman and Sharpe (2003) later imported multiple imputation into empirical finance, showing on several canonical applications that complete-case analysis materially biases the conclusions.

The topic has recently returned to the centre of asset pricing, driven by the firm-characteristics panels used in cross-sectional factor models, where the majority of entries can be missing. Bryzgalova et al. (2025) document that the profession's ad hoc fixes, typically cross-sectional medians, are not innocuous: the choice of imputation changes the estimated risk premia, and they propose a factor-based imputation exploiting both cross-sectional and time series dependence. In parallel, Freyberger et al. (2025) show that firm characteristics are missing *not* at random and develop estimators that remain consistent under such mechanisms. Both papers make, in the cross-section of characteristics, the point this thesis makes for return panels: what matters is not the imputation error itself but its effect on the downstream object of interest. Finally, Choudhury et al. (2021) highlight a specifically financial pitfall: imputing with a model trained on the full sample leaks future information into the imputed values, and they formalize the resulting trade-off between look-ahead bias and imputation variance.

Summing up, the statistical literature provides the taxonomy of mechanisms and the classical estimators, the machine learning literature provides increasingly powerful conditional models trained under an implicit MAR assumption, and the finance literature provides both the warning that missingness is often MNAR precisely when it matters most (crises, illiquidity) and the reminder that imputation should be judged by the downstream model. The experimental design of this thesis, controlled injection of MCAR and MNAR patterns into a complete equity panel, classical and deep imputers side by side, and evaluation on covariance geometry, risk measures and portfolio construction rather than pointwise error alone, sits deliberately at the intersection of these three strands.

1.3 Research questions

The thesis is organised around four questions.

1. **What drives the damage, the amount of missing data or the mechanism that produced it?** We compare patterns that remove entries at random with a pattern that removes them preferentially in turbulent periods, at comparable shares of missing cells.

2. **Do the deep imputers earn their cost?** Recurrent and diffusion-based models are the current state of the art on imputation benchmarks. We ask whether they improve on classical likelihood and low-rank estimators on a financial panel, and what any improvement is worth once training and sampling time are counted.

3. **Is pointwise accuracy a good proxy for the quality of the imputed panel?** Reconstruction error is the standard criterion, but no downstream model consumes individual cells. We ask whether a method that fills the holes accurately also preserves the covariance matrix, the tails, the marginal distributions and the temporal dependence of the returns.

4. **Do the rankings survive downstream?** We ask whether the ordering of the methods given by the reconstruction metrics is the same ordering given by the decisions a user would take on the imputed panel, and where the two disagree.

1.4 Contributions

1. A controlled testbed for imputation on financial data: a complete panel of daily log returns for fifty S&P 500 constituents, three injection mechanisms at five severities each, and seven imputers from deletion to a conditional diffusion model, all implemented and run within a single framework.

2. An evaluation suite that goes past reconstruction error. Alongside MAE and RMSE we score the covariance matrix, the five percent Value-at-Risk, the marginal distributions in Wasserstein distance, and two temporal statistics, the autocorrelation of absolute returns and the leverage curve, which the permutation-invariant metrics cannot detect.

3. The downstream evaluations, in which an imputation is judged by the object a user would build on it, referenced against the same object built on the uncorrupted panel. We instantiate the idea twice, with the tangency portfolio and with the factor structure of the panel, and design both to rank the reconstruction methods against one another rather than against a naive baseline.

4. The empirical findings themselves, among them that pointwise and structural quality can diverge by close to an order of magnitude, that the neural models do not improve on EM at a computational cost several orders of magnitude higher, that every conditional imputer inflates the factor structure of the panel because the values it inserts are conditional means, and that the ranking of the methods is least stable exactly where the data is missing not at random.

1.5 Outline of the thesis

Chapter 2 describes the data, the three injection mechanisms, the seven imputation models and the evaluation criteria, both the pointwise and structural metrics and the two downstream evaluations. Chapter 3 reports the results, one table per injection pattern, then the portfolio and the factor-structure evaluations, and closes with the computational cost of each method. Chapter 4 returns to the four questions above and states what remains open.

Chapter 2

Methodology

As already mentioned, missing data can arise in countless different ways, and the choice of methodology depends on the nature of the missingness, the type of data, and the specific goals of the analysis. We will focus on a subset of cases, namely those that involve imputation of the missing values with "information" from the whole dataset (no causal constraint), with in mind the goal of training a downstream model that is as close as possible to the one that would have been trained on the complete dataset. Our choice of data, models, and evaluation methods will be guided by these constraints. We will nevertheless mention when/how models should be tweaked to solve adjacent problems.

2.1 Data

As our main baseline, we chose the most studied and reliable dataset in finance: the end-of-day prices of the S&P500, in this case fetched with yfinance from 1/1/2010 to 30/5/2026 (potrei cambiare le date). It clearly is not a dataset inflicted with the curse of missing data; the only structural ones being occasional circuit breaks and listings/delistings from the index. To obtain a full dataset with no "real" missing data, we removed each stock that contained at least one missing observation. Furthermore, we selected a sample of 50 stocks from the index, stratified across industries; this is a sensible choice because:

1. with more 400 stocks the covariance matrix $\mathbf{\Sigma}$ quickly becomes unreliable (noisy covariances, skewed eigenvalues, unstable inversion...) as the number of observations shrinks;

2. it is easier to check whether the ground data is actually correct;

3. model training requires considerably less memory, in a time where RAM and GPU prices are skyrocketing;

4. it is well known that a well constructed portfolio with more than 30 stocks is already sufficiently diversified (since volatility scales with the inverse of the square root of

the number of securities).

This dataset will serve as the ground truth against which we will compare our results. In parallel, we will modify it by inserting fake missing observations according to different rules, as we define in the subsequent sections.

Now an important choice is whether to operate in the prices "space" P_t or the log returns space R_t. There are positives and negatives about each. For one, the raw dataset is prices, and in practice missing data will mostly come in the form of missing prices. Furthermore, in most applications we care slightly more about prices than returns, since the formers are what is actually bought and sold. But P_t has the drawback of not being a complete space ($p_{t,k} > 0$) and being highly level dependant, making the series non-stationary and very difficult for ML algorithms to handle. We will thus focus on R_t.

2.1.1 Missingness Injection

(Da scrivere meglio, questa è solo l'outline)

As outlined in the literature review ...

We used three algorithms to simulate different missingness patterns:

1. **Sparse**: each scalar $r_{t,k}$ is missing with probability r, independent of the rest of the dataset. This is the easiest example of MCAR and can simulate a randomly corrupted dataset;

2. **Contiguous gaps**: n points $r_{t,k}$ are chosen at random and eliminated. Then subsequent observations of the same variable $r_{t+1,k}, \ldots, r_{t+h,k}$ get deleted according to an exponential decay with rate λ. This is still an example of MCAR and can be thought to simulate assets failing to trade because of random shortages of liquidity;

3. **Stressed**: we calculate index squared returns for each date t and rank them. Then, returns whose date is above a certain percentile $p$get deleted with probability h, while the rest still go missing with rate r. This simulates shortages of liquidity or circuit breaks that happen when markets experience high levels of volatility; since the missingness of a given $r_{t,k}$ is injected conditional on all the other values of the dataset, this is an example of MNAR.

For each of the three missingness patterns, we will vary the parameters r, λ, and h to simulate different levels of missingness. Furthermore, we will make it so that the ratio of missing values is the same for each level of missingness in each pattern, so that we can compare the performance of the different imputation methods in a sensible way. The exact values of the parameters are reported in Table 2.2. The column "Rate" reports the expected value of the missingness ratio for each pattern.

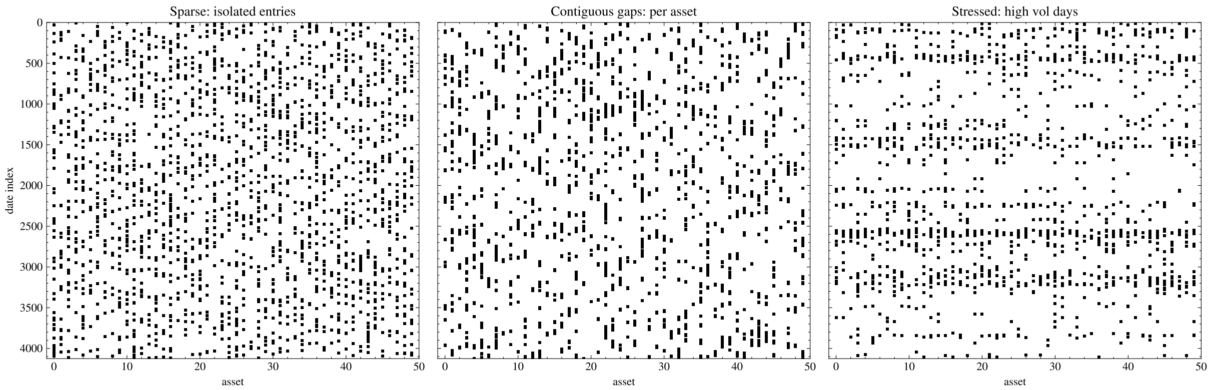

Figure 2.1: Missingness Examples

Sparses (r)	**Gaps** (n, λ)	**Stressed** (r, h, p)	**Rate**
0.001	(20, 0.1)	(0, 0.1, 0.99)	0.001
0.01	(200, 0.1)	(0.005, 0.5, 0.99)	0.01
0.05	(1000, 0.1)	(0.04, 0.2, 0.95)	0.05
0.1	(1000, 0.05)	(0.05, 0.5, 0.9)	0.1
0.3	(2000, 0.03)	(0.2, 1, 0.9)	0.3

Figure 2.2: Missingness parameters

2.2 Models

2.2.1 Deletion

The baseline of complete-case analysis: every date containing at least one missing value is dropped, and all downstream estimation runs on the surviving complete rows. Under MCAR the retained dates are a random subsample of the panel, so moment estimates remain consistent (Little and Rubin, 2019); the cost is statistical efficiency, and in the cross-section it is severe. With sparse missingness at rate r, a date survives with probability $(1-r)^N$: at $r = 0.05$ and $N = 50$ assets, fewer than 8% of the dates survive. Under the stressed pattern the damage changes nature rather than size: the deleted dates are exactly the turbulent ones, so the surviving subsample is calm by construction and every risk measure computed from it is biased downwards, no matter how much data is left.

2.2.2 Mean Replacement

The simplest imputation that preserves the shape of the panel consists in setting each missing return to its unconditional expected value. This is almost equivalent to setting the missing price to be equal to the last observed price, since daily returns are very close to zero. The mean of the panel is preserved, but every second moment is damped: under sparse missingness at rate r, a variance term survives whenever its entry is observed while a covariance term needs both entries at once, so estimated volatilities scale by $\sqrt{1-r}$ and pairwise correlations by $(1-r)$, shrinking the covariance matrix towards its diagonal. Under the stressed pattern the distortion is stronger still, because the entries being replaced are on average the largest ones. This predictable failure is precisely what the structural metrics of Section 2.3 are designed to expose.

2.2.3 Low Rank Matrix Completion

A few pervasive factors describe most of the covariance of equity returns, so the return panel is close to a low-rank matrix plus idiosyncratic noise. Matrix completion turns this observation into an imputation device: if the underlying matrix has rank $q \ll N$, the missing entries can be recovered from the observed ones – exactly so, under conditions on the missingness pattern and on how spread out the factors are (Candès and Recht, 2009). We adopt the iterative-SVD scheme of Mazumder et al. (2010): standardise each asset on its observed entries, fill the missing cells with zeros, then alternate between projecting the completed panel onto the set of rank-q matrices (truncated SVD) and restoring the observed entries, until the panel stops changing. The fixed point fills every missing cell with a q-factor reconstruction estimated on the fly, without ever specifying the factors. The nuclear-norm variant of the same algorithm, which soft-thresholds the singular values instead of truncating them, shrinks every imputed entry towards zero and thereby inherits a milder version of the covariance damping of mean replacement; we therefore keep the hard rank constraint, with q small, in line with the number of pervasive equity factors.

2.2.4 EM

Our likelihood-based model treats the dates as i.i.d. draws $r_t \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and maximises the resulting observed-data likelihood with the EM algorithm of Dempster et al. (1977). Write $r_t = (r_{t,o}, r_{t,m})$ for the observed and missing parts of date t (the split changes from date to date). The E-step fills each missing block with its conditional expectation,

$$\mathbb{E}\left[r_{t,m} \mid r_{t,o}\right] = \boldsymbol{\mu}_m + \boldsymbol{\Sigma}_{mo}\, \boldsymbol{\Sigma}_{oo}^{-1} \left(r_{t,o} - \boldsymbol{\mu}_o\right),$$

while the M-step re-estimates $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ from the completed panel, adding back the conditional covariance $\boldsymbol{\Sigma}_{mm} - \boldsymbol{\Sigma}_{mo}\boldsymbol{\Sigma}_{oo}^{-1}\boldsymbol{\Sigma}_{om}$ of the cells just filled – the correction that separates EM from naively iterating a regression fill, which would understate the covariance. At convergence we obtain the maximum likelihood estimate $(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}})$ and, as a by-product, imputations that are exactly the L^2 projections discussed in Section **??**. In the implementation, dates sharing the same missingness pattern are grouped together, so each pattern costs a single linear solve per iteration. Daily returns are of course heavier-tailed than the Gaussian assumption wants; the same iteration extends to a multivariate Student-t, whose E-step gains a per-date weight, decreasing in the Mahalanobis distance of the observed block, that discounts extreme dates in the update of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ (Little, 1988; Liu and Rubin, 1995).

2.2.5 KNN

k-nearest-neighbours imputation, first popularised on gene-expression data by Troyanskaya et al. (2001), is the donor-based, fully non-parametric member of our set. To fill date t, its distance to every other date is computed as the Euclidean distance over the assets both dates observe, rescaled by the number of co-observed assets so that sparser rows are not mechanically favoured; the missing return $r_{t,i}$ is then the inverse-distance-weighted average of $r_{s,i}$ over the k closest dates s that do observe asset i. Nothing about the scheme is temporal: donors are selected by cross-sectional similarity, wherever they sit in the sample. No distributional assumption is needed, but the method degrades exactly where donors become unreliable: under heavy missingness the distances rest on few co-observed assets, and under the stressed pattern the extreme dates that most need filling have no close, well-observed counterparts to borrow from.

2.2.6 BRITS

BRITS (Cao et al., 2018) is our recurrent neural representative. A window of the panel is processed by two LSTMs, one per time direction. At each step the hidden state is first decayed by a factor learned from the time elapsed since each asset was last observed, so that the memory of a stale reading fades, and two candidate fills are produced: a history-based estimate regressed from the hidden state, and a cross-sectional estimate regressed from the other assets at the same date. A learned gate blends the two, the blend replaces the missing entries, and the completed vector is fed to the next step, so the imputations themselves carry gradients through the network. Training is self-supervised: the network learns to reconstruct the entries that are observed, with a penalty tying the forward and backward passes together, and the missing cells are then read out of the trained network. No explicit model of the returns or of the missingness mechanism is assumed.

2.2.7 CSDI

CSDI (Tashiro et al., 2021) adapts denoising diffusion models (Ho et al., 2020) to imputation. In a diffusion model, data is progressively corrupted by Gaussian noise over many small steps until only noise remains, and a network is trained to undo one corruption step at a time; generation then amounts to running the learned chain backwards from pure noise. CSDI makes this chain conditional: the denoising network – a transformer applying attention along time and across assets (Vaswani et al., 2017) – always sees the clean observed entries while denoising the missing ones. Training is self-supervised, hiding a random share of the observed cells and learning to denoise them given the rest. At inference the chain starts from noise on the missing cells, with the observed ones held fixed throughout, and each pass produces one draw from the predictive distribution of the missing block given the observed panel; the final imputation is the median over several draws. CSDI is thus the one genuinely probabilistic model of our set – the dispersion of its draws is a built-in estimate of the imputation uncertainty, which no other model here provides.

2.3 Evaluation Methods

Every experiment produces, for a given missingness pattern and level, an imputed panel $\widehat{R} \in \mathbb{R}^{T \times N}$ to be compared against the ground truth R of Section 2.1. Let $\mathcal{M} = \{(t,k) : r_{t,k} \text{ was deleted}\}$ denote the set of imputed cells and $|\mathcal{M}|$ its cardinality. Since our stated goal is not the recovery of individual values but of the panel a downstream model would have been trained on, we score the imputations with two families of metrics. *Pointwise* metrics are computed on the imputed cells alone – the observed cells are returned untouched by every model, and including them would only dilute the error – and measure how close each filled value is to the deleted one. *Structural* metrics instead compare statistics computed from the two full panels, and measure whether the objects that downstream models actually consume (covariances, tail quantiles, marginal distributions, temporal dependence) survive the imputation. The two families need not agree: a method can be nearly optimal pointwise and still destroy the second-moment structure, as mean replacement shows. Deletion is the exception in what follows: it returns a panel with fewer rows, so the pointwise metrics are not defined for it and the structural ones are computed on the surviving subsample.

1. **MAE**: the mean absolute error on the imputed cells,

$$\mathrm{MAE} = \frac{1}{|\mathcal{M}|} \sum_{(t,k)\in\mathcal{M}} \left|\hat{r}_{t,k} - r_{t,k}\right|,$$

   expressed in return units and robust to occasional large misses.

2. **RMSE**: the root mean squared error on the same cells,

$$\mathrm{RMSE} = \left(\frac{1}{|\mathcal{M}|} \sum_{(t,k)\in\mathcal{M}} \left(\hat{r}_{t,k} - r_{t,k}\right)^2 \right)^{1/2}.$$

   This is the sample counterpart of the L^2 distance of Section **??**: the imputation minimising its expectation is the conditional expectation given the observed data, so RMSE rewards exactly the projections that the likelihood-based models construct, while penalising large errors quadratically.

3. **Covariance error**: the relative Frobenius error of the sample covariance matrix,

$$\mathrm{CovErr} = \frac{\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_F}{\|\boldsymbol{\Sigma}\|_F},$$

   where $\boldsymbol{\Sigma}$ and $\widehat{\boldsymbol{\Sigma}}$ are estimated from the true and the imputed panel respectively. This is our central structural metric: the covariance matrix is the object whose geometry (Section **??**) we set out to preserve, and the input of essentially every allocation and risk model built downstream.

4. **Value-at-Risk error**: for each asset the empirical VaR at level $\alpha = 0.05$ is the (sign-flipped) empirical α-quantile of its return series, $\mathrm{VaR}_\alpha(k) = -\hat{q}_\alpha(R_k)$, and we report

$$\mathrm{VaRErr} = \frac{1}{N} \sum_{k=1}^{N} \left| \widehat{\mathrm{VaR}}_\alpha(k) - \mathrm{VaR}_\alpha(k) \right|.$$

   This probes the left tail beyond the second moment, the region that risk management actually regulates – and precisely the region that the stressed pattern of Section 2.1.1 empties out.

5. **Wasserstein error**: the mean 1-Wasserstein distance between the per-asset marginal return distributions,

$$\text{WassErr} = \frac{1}{N}\sum_{k=1}^{N} W_1\left(\widehat{F}_k, F_k\right), \qquad W_1(F,G) = \int_{\mathbb{R}} |F(x) - G(x)|\,dx,$$

with F_k and $\widehat{F}_k$ the empirical distribution functions of asset k in the true and imputed panel. W_1 measures the minimal average displacement needed to transport one distribution onto the other; it compares the marginals in full, in return units, and detects shape distortions – e.g. imputed values overly concentrated near zero – that moment-based metrics summarise away.

6. **Absolute-return ACF error**: every metric above is invariant to a permutation of the dates; the last two are not. Daily returns are themselves close to serially uncorrelated, but their absolute values are persistently and positively autocorrelated – volatility clustering, the most robust temporal stylized fact of asset returns (Cont, 2001). Writing $\hat{\rho}_k(\cdot)$ for the sample autocorrelation at lag k, we compare the autocorrelation functions of the absolute returns up to a horizon K,

$$\text{AbsACFErr} = \frac{1}{NK}\sum_{j=1}^{N}\sum_{k=1}^{K}\left|\hat{\rho}_k\left(|\widehat{R}_j|\right) - \hat{\rho}_k(|R_j|)\right|,$$

with $K = 20$ throughout. An imputer that fills the panel with values that are individually plausible but temporally independent – as any model that treats the dates as exchangeable must – erases this dependence and is caught here and nowhere above.

7. **Leverage error**: negative returns raise subsequent volatility more than positive ones of the same size, the leverage effect (Black, 1976): the cross-correlation curve

$$L_k = \text{corr}(r_t, |r_{t+k}|), \qquad k = 1, \ldots, K,$$

is negative at short lags. We report the mean absolute gap between the true and imputed curves, averaged over assets and lags as above. Unlike the previous metric, which is blind to sign, this one probes an *asymmetric* dependence between the level of a return and the volatility that follows it, and thereby whether the imputed panel still distinguishes falling from rising markets.

All seven metrics are computed by a single evaluation routine for every model, missingness pattern and missingness level; the resulting tables are collected and discussed in

Chapter 3.

2.3.1 Downstream evaluations

The seven metrics above compare the imputed panel with the truth directly. A complementary and in some ways more honest test asks a different question: if we build several models on the imputed panel, how far is their output from each other? We call this class of criteria downstream evaluations. We fix a downstream model, fit it once on each imputed panel, and compare the fitted objects across the imputation methods and against the one fitted on the uncorrupted panel. The aim is to rank the imputation methods against one another rather than against a naive baseline, so we keep these results separate from the tables of Chapter 3. Two such evaluations follow, the optimal portfolio and the factor structure of the panel.

Optimal portfolio

The most classical downstream model in finance is mean-variance portfolio selection (Markowitz, 1952). From a return panel we estimate the mean vector $\hat{\boldsymbol{\mu}}$ and the covariance matrix $\hat{\boldsymbol{\Sigma}}$ and form the fully invested tangency portfolio, the point of the efficient frontier with the highest ratio of expected return to volatility:

$$\hat{w} = \frac{\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\mu}}}{\mathbf{1}^\top\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\mu}}}, \qquad \mathbf{1}^\top\hat{w} = 1.$$

Short positions are allowed, as in the unconstrained classical problem. The covariance enters through its inverse, so any distortion the imputation leaves in $\hat{\boldsymbol{\Sigma}}$, the very object Section **??** set out to preserve, is carried into the weights and often amplified there. The portfolio is in this sense a direct stress test of the structural quality that the covariance error of Section 2.3 measures in the aggregate.

We compute $\hat{w}$ once for every imputed panel and once for the uncorrupted panel. To judge an allocation we do not trust the panel it was estimated on: we realise each weight vector on the true returns,

$$r_t^p = r_t^\top\hat{w},$$

and report the annualised mean and variance of the resulting series r^p, together with its Sharpe ratio, annualising with 252 trading days and taking a zero risk-free rate. The portfolio built on the true panel is the oracle: it is the best in-sample allocation available, and every imputation is judged by how close its realised moments, and its weights, come to it. A method that preserves the second-moment geometry recovers weights near the oracle and a comparable realised mean and variance. One that distorts that geometry, or that discards data as complete-case deletion does, produces weights that look optimal on the panel it saw and perform poorly once confronted with the real returns. The comparison is reported in Chapter 3, in a table of realised moments and a plot of the realised series, kept apart from the metric tables above.

Factor structure

The portfolio reads the covariance through its inverse, so it measures the distortion left by the imputation and the amplification of that distortion at the same time. The second downstream model separates them. We take the statistical factor model of the panel, that is the principal components of $\hat{\boldsymbol{\Sigma}}$, and compare the factors extracted from each imputed panel with those of the uncorrupted one. Nothing is inverted here, so what the comparison reports is the damage to the second moments alone.

Write the spectral decomposition of the estimated covariance as

$$\hat{\boldsymbol{\Sigma}} = \sum_{i=1}^{N} \hat{\lambda}_i \, \hat{v}_i \hat{v}_i^\top, \qquad \hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_N \geq 0,$$

with orthonormal eigenvectors $\hat{v}_i$. The first k of them, collected in the $N \times k$ matrix $\hat{V}_k$, are the loadings of the k leading statistical factors, and in a panel of equity returns the first of these is the market factor. We report two things. The first is the spectrum, through the share of the total variance

$$\hat{\lambda}_i \Big/ \sum_{j=1}^{N} \hat{\lambda}_j$$

carried by each component, which says whether the imputation has concentrated the panel into fewer factors than the data support or spread it over more. The second is the position of the factor space itself. Let V_k be the true loadings and let $\theta_1 \leq \cdots \leq \theta_k$ be the principal angles between the two subspaces, the cosines of which are the singular values of $V_k^\top \hat{V}_k$. We summarise them by

$$D_k = \frac{1}{\sqrt{k}} \left(\sum_{i=1}^{k} \sin^2 \theta_i \right)^{1/2} \in [0, 1],$$

the root mean square of their sines, which is 0 when the two factor spaces coincide and 1 when they are orthogonal. Comparing subspaces rather than individual eigenvectors makes the criterion invariant to the sign of each $\hat{v}_i$ and to any reordering among factors of nearly equal variance, both of which are arbitrary. Dividing by $\sqrt{k}$ keeps the values comparable across different numbers of factors.

The two downstream evaluations are connected by the same decomposition. Since $\hat{\boldsymbol{\Sigma}}^{-1} = \sum_i \hat{\lambda}_i^{-1} \hat{v}_i \hat{v}_i^\top$, the inverse on which the portfolio of Section 2.3.1 depends weights each direction by the reciprocal of its variance, so it is dominated by the eigenvectors carrying the smallest eigenvalues. These are the directions supported by the least data, and the ones an imputation has the least information to recover. Reporting D_k as k grows therefore says how deep into the spectrum an imputation remains trustworthy, which is what the portfolio implicitly relies on. We report D_k at $k = 1$ and $k = 5$ together with the variance shares, and plot the full spectrum and angle profile, in Chapter 3.

Chapter 3

Results and Discussion

Tables 3.1, 3.2 and 3.3 report the seven evaluation metrics of Section 2.3 for every model, one table per injection pattern of Section 2.1.1. Within each table the levels are ordered by their realized share of missing cells, and within each level the best value of every metric is set in bold. A dash marks a value that is not defined. This happens for the pointwise errors of Deletion, since the panel it returns changes shape, for its remaining metrics at the heaviest levels, where too few complete dates survive, and for KNN at the most extreme stressed level, where the dates to be filled have no well-observed donors left.

The ranking of the models is stable across patterns and levels. The conditional methods dominate the two baselines everywhere, EM is rarely beaten and never by a wide margin, and the neural models match it at a much higher computational cost rather than improving on it; BRITS falls clearly behind at two of the stressed levels.

That EM is so hard to beat is less surprising than it first looks. Daily returns are close to serially uncorrelated, so nearly everything that can be known about a missing entry is carried by the returns observed on the same date, and the conditional mean of Section 2.2 extracts exactly that. What is left over is idiosyncratic and no model can predict it. The pointwise columns therefore have a floor, and it is reached almost at once: EM's MAE stays between 0.0071 and 0.0080 across the whole sparse table, and the five conditional methods are separated in the third decimal. The additional capacity of BRITS and CSDI has little to act on, since the expectation of a return given its cross-section is close to linear and the residual is close to noise. They are not punished for that capacity, they simply cannot be paid for it.

The same floor is why the pointwise and the structural columns disagree about severity. MAE and RMSE average over the imputed cells, so they report how hard a typical fill is and stay flat as the number of fills grows: EM's sparse MAE is 0.0080 at rate 0.001 and 0.0074 at rate 0.3, and zero imputation is just as flat at about 0.011 throughout. The covariance error accumulates those same per-cell errors over the whole panel, and across the same range it grows by a factor of almost forty, from 0.0015 to 0.0565. Read down the two families, the amount of missing data is nearly invisible in the first and is the leading term in the second. This is the distinction of Section 2.3 between pointwise and structural quality in its plainest form: under the MCAR patterns the MAE of zero imputation is only about fifty percent larger than the best one, while its covariance error is close to an order of magnitude worse than that of EM, so a model can look acceptable entry by entry and still distort the geometry of the panel.

The two families do not even agree on what a good fill is. Minimising squared error asks for a conditional mean, which is by construction less variable than the return it replaces, so the fill that is best entry by entry is also the one that damps the second moment most. The tables show the exchange being made. At the stressed levels (0.005, 0.5, 0.99) and (0.05, 0.5, 0.9) low-rank completion loses to EM on MAE and beats it on the covariance error, and at sparse rate 0.3 BRITS does the same. Low-rank completion is helped here by the norm as much as by the fill: a relative Frobenius error weighs the large entries of $\mathbf{\Sigma}$, which are the factor-driven ones, so a method that places every fill on the leading subspace reproduces those and neglects the directions the norm barely counts. Section 3.2 shows what that costs.

Which metric is read also decides how much can be told apart at all. The Value-at-Risk and Wasserstein errors are computed one asset at a time, and a marginal distribution of daily returns is easy to reproduce: a fill of roughly the right size leaves it almost intact, which is why both columns are zero to four decimal places at the lighter levels and separate the models only at the heaviest. At sparse rate 0.3 zero imputation is two and a half times worse than the best model on the Value-at-Risk error and nearly nine times worse on the covariance error, from the same panel. The covariance, absolute-return autocorrelation and leverage errors are all built from products of returns, and they require a fill to stand in the right relation to the other assets on its date and to the neighbouring dates. Producing a plausible number is easy; producing a number that is plausible jointly with the rest of the panel is not, and only the second family of metrics can tell the two apart.

The three patterns then differ in which of those relations they destroy. Since the levels are matched on the share of missing cells, the row blocks of the three tables can be read across, and the first result of doing so is that the two MCAR tables are almost interchangeable. EM's covariance error under contiguous gaps is never more than five percent worse than under sparse missingness and at three of the five levels it is better, although the two patterns delete cells in entirely different geometries, isolated entries in one case and runs of up to thirty consecutive days in the other. Breaking the time axis costs a cross-sectional imputer nothing, because on any date the other forty-nine assets are still observed, and for a nearly serially uncorrelated return that is the informative conditioning set anyway. The only model with anything at stake is BRITS, which propagates along time, and the gap table shows both sides of it: at (20, 10) the runs are short, the decayed history is still fresh next to each hole and BRITS has the best MAE and RMSE at that level, while at (2000, 30) that channel has faded and it falls back on the cross-section, ending up indistinguishable from EM.

The stressed pattern removes the conditioning set itself. Its deletions are drawn by date and are therefore aligned across assets, so the cross-section that the other two patterns leave intact is exactly what thins out, to the point where KNN has no well-observed donors left at the last level and returns nothing at all. At matched shares the gap is wide: at about ten percent missing EM's covariance error is 0.0237 under sparse, 0.0238 under gaps and 0.0554 under stressed, and at about thirty percent it is 0.0565, 0.0589 and 0.6634. The thinner conditioning is the smaller half of the explanation. The deleted returns are not a random draw but the largest ones, so a model fitted to the calm surviving sample fills turbulent dates with calm values and has nothing to tell it otherwise. The error sits in the target rather than in the estimator, which is why the stressed table degrades in a way that no amount of model capacity addresses. At the heaviest level no method brings the relative covariance error below 0.6; the turbulent dates are simply no longer in the data, and nothing in the observed sample allows any model to recover them.

This is also what separates the two levels at which BRITS falls behind, (0, 0.1, 0.99) and (0.05, 0.5, 0.9), the two whose deleted entries are the most extreme relative to what stays visible. In the first, every missing cell lies on the one percent most turbulent dates, and BRITS returns an MAE of 0.0301 against EM's 0.0133, behind even KNN. EM's fill is a linear function of the block observed on the same date, so when the other assets move by five percent the fill moves with them, whereas a regression head trained self-supervised on a panel of ordinary dates has learned to produce ordinary values and does not extrapolate. In the second, half of the entries on the most turbulent tenth of dates are gone, which degrades both of BRITS's channels at once, and its covariance error is nearly four times EM's.

The largest numbers in the tables belong to Deletion, and they measure something other than imputation error. It imputes nothing, so its covariance error of 1.4799 at (2000, 30), more than three times that of zero imputation, records how few dates survive the requirement that all fifty assets be observed at once, and under the stressed pattern it records in addition that the survivors are calm by construction.

At the extreme stressed level, finally, the ranking compresses to nothing. Every surviving model reports an MAE between 0.0134 and 0.0137 and a covariance error between 0.648 and 0.663, and zero imputation, hopeless everywhere else, reaches 0.768. The differences between imputation methods are informative only while there is something left in the panel to recover.

Table 3.1: Evaluation scores under sparse missingness (MCAR); levels are the deletion rate r.

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
0.001	Deletion	–	–	0.0119	0.0001	0.0001	0.0101	0.0114
	Zero	0.0118	0.0194	0.0026	**0.0000**	**0.0000**	0.0009	0.0008
	KNN	0.0099	0.0168	0.0018	**0.0000**	**0.0000**	0.0006	0.0005
	LowRank	0.0084	0.0152	0.0016	**0.0000**	**0.0000**	**0.0005**	**0.0004**
	EM	**0.0080**	**0.0147**	**0.0015**	**0.0000**	**0.0000**	**0.0005**	**0.0004**
	BRITS	0.0091	0.0158	0.0016	**0.0000**	**0.0000**	**0.0005**	**0.0004**
	CSDI	0.0081	0.0151	0.0016	**0.0000**	**0.0000**	**0.0005**	**0.0004**
0.01	Deletion	–	–	0.0532	0.0005	0.0004	0.0247	0.0334
	Zero	0.0107	0.0161	0.0205	0.0001	0.0001	0.0035	0.0027
	KNN	0.0081	0.0123	0.0069	0.0001	**0.0000**	0.0019	0.0017
	LowRank	0.0077	0.0117	0.0054	**0.0000**	**0.0000**	0.0016	0.0015
	EM	0.0071	0.0109	0.0051	**0.0000**	**0.0000**	**0.0015**	**0.0014**
	BRITS	0.0076	0.0116	0.0064	0.0001	**0.0000**	0.0017	0.0016
	CSDI	**0.0069**	**0.0107**	**0.0050**	**0.0000**	**0.0000**	**0.0015**	**0.0014**
0.05	Deletion	–	–	0.1951	0.0019	0.0012	0.1334	0.0515
	Zero	0.0111	0.0167	0.1044	0.0005	0.0006	0.0132	0.0073
	KNN	0.0084	0.0127	0.0283	0.0003	0.0002	0.0047	0.0045
	LowRank	0.0078	0.0118	0.0169	**0.0002**	**0.0001**	0.0045	0.0041
	EM	**0.0072**	0.0113	0.0151	**0.0002**	0.0002	0.0044	**0.0039**
	BRITS	0.0073	0.0113	**0.0145**	**0.0002**	0.0002	**0.0043**	0.0040
	CSDI	**0.0072**	**0.0112**	0.0149	**0.0002**	0.0002	0.0051	0.0041
0.1	Deletion	–	–	0.4492	0.0091	0.0051	–	–
	Zero	0.0109	0.0163	0.1872	0.0011	0.0011	0.0276	0.0111
	KNN	0.0083	0.0125	0.0487	0.0007	0.0004	0.0090	0.0067
	LowRank	0.0077	0.0116	0.0282	**0.0004**	**0.0003**	0.0072	0.0058
	EM	**0.0071**	**0.0110**	**0.0237**	0.0005	**0.0003**	**0.0068**	**0.0056**
	BRITS	0.0072	0.0111	0.0238	0.0005	**0.0003**	**0.0068**	0.0057
	CSDI	0.0072	0.0111	0.0261	**0.0004**	**0.0003**	0.0077	0.0057
0.3	Deletion	–	–	–	–	–	–	–
	Zero	0.0109	0.0161	0.4749	0.0037	0.0033	0.0638	0.0201
	KNN	0.0087	0.0129	0.1467	0.0024	0.0014	0.0173	0.0118
	LowRank	0.0080	0.0119	0.0749	**0.0014**	**0.0007**	0.0170	0.0105
	EM	**0.0074**	**0.0112**	0.0565	0.0017	0.0009	0.0150	**0.0099**
	BRITS	0.0076	0.0114	**0.0536**	0.0015	0.0008	**0.0149**	0.0101
	CSDI	**0.0074**	0.0113	0.0641	**0.0014**	0.0008	0.0203	0.0112

Table 3.2: Evaluation scores under contiguous-gap missingness (MCAR); levels are the parameters (n, λ) of Section 2.1.1.

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(20, 10)	Deletion	–	–	0.0103	0.0001	0.0001	0.0033	0.0033
	Zero	0.0121	0.0184	0.0027	**0.0000**	**0.0000**	0.0003	0.0002
	KNN	0.0096	0.0144	0.0016	**0.0000**	**0.0000**	0.0002	**0.0001**
	LowRank	0.0087	0.0123	0.0010	**0.0000**	**0.0000**	0.0002	**0.0001**
	EM	0.0079	0.0117	0.0011	**0.0000**	**0.0000**	**0.0001**	**0.0001**
	BRITS	**0.0078**	**0.0112**	0.0010	**0.0000**	**0.0000**	**0.0001**	**0.0001**
	CSDI	0.0079	0.0115	**0.0009**	**0.0000**	**0.0000**	**0.0001**	**0.0001**
(200, 10)	Deletion	–	–	0.1492	0.0010	0.0004	0.0243	0.0158
	Zero	0.0107	0.0156	0.0222	**0.0001**	0.0001	0.0037	0.0015
	KNN	0.0082	0.0121	0.0065	**0.0001**	**0.0000**	0.0016	0.0012
	LowRank	0.0078	0.0118	0.0061	**0.0001**	**0.0000**	0.0015	0.0012
	EM	**0.0072**	**0.0109**	**0.0046**	**0.0001**	**0.0000**	**0.0014**	**0.0011**
	BRITS	0.0073	0.0111	0.0049	**0.0001**	**0.0000**	0.0015	**0.0011**
	CSDI	**0.0072**	**0.0109**	0.0049	**0.0001**	**0.0000**	0.0015	**0.0011**
(1000, 10)	Deletion	–	–	0.1825	0.0017	0.0010	0.1135	0.0493
	Zero	0.0104	0.0152	0.0785	0.0004	0.0005	0.0155	0.0048
	KNN	0.0080	0.0118	0.0224	0.0003	0.0002	0.0059	0.0032
	LowRank	0.0074	0.0108	0.0137	**0.0002**	**0.0001**	**0.0048**	0.0029
	EM	**0.0069**	**0.0103**	0.0116	**0.0002**	**0.0001**	0.0049	0.0028
	BRITS	0.0070	0.0104	0.0116	**0.0002**	**0.0001**	0.0049	0.0028
	CSDI	**0.0069**	**0.0103**	**0.0114**	**0.0002**	**0.0001**	0.0051	**0.0027**
(1000, 20)	Deletion	–	–	0.8524	0.0087	0.0042	0.1872	0.1681
	Zero	0.0110	0.0163	0.1716	0.0010	0.0010	0.0354	0.0068
	KNN	0.0085	0.0127	0.0487	0.0007	0.0004	0.0117	0.0049
	LowRank	0.0078	0.0119	0.0310	**0.0004**	**0.0003**	0.0088	0.0046
	EM	**0.0073**	**0.0114**	**0.0238**	0.0005	**0.0003**	0.0083	**0.0044**
	BRITS	0.0074	**0.0114**	0.0241	0.0005	**0.0003**	**0.0082**	0.0045
	CSDI	**0.0073**	0.0115	0.0328	**0.0004**	**0.0003**	0.0095	**0.0044**
(2000, 30)	Deletion	–	–	1.4799	0.0104	0.0100	–	–
	Zero	0.0108	0.0161	0.4324	0.0028	0.0027	0.0908	0.0143
	KNN	0.0085	0.0128	0.1378	0.0018	0.0011	0.0277	0.0090
	LowRank	0.0080	0.0120	0.0820	**0.0010**	**0.0006**	0.0218	0.0083
	EM	**0.0073**	**0.0113**	0.0589	0.0012	0.0007	0.0212	**0.0079**
	BRITS	0.0074	0.0114	**0.0555**	0.0011	**0.0006**	**0.0187**	0.0082
	CSDI	0.0074	0.0115	0.0896	**0.0010**	0.0007	0.0265	0.0091

Table 3.3: Evaluation scores under stressed missingness (MNAR); levels are the parameters (r, h, p) of Section 2.1.1.

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(0, 0.1, 0.99)	Deletion	–	–	0.2813	0.0009	0.0004	0.0582	0.0200
	Zero	0.0552	0.0640	0.0676	0.0001	0.0001	0.0115	0.0070
	KNN	0.0202	0.0285	0.0229	**0.0000**	**0.0000**	0.0034	0.0029
	LowRank	0.0158	0.0220	0.0116	**0.0000**	**0.0000**	0.0025	0.0020
	EM	**0.0133**	**0.0187**	**0.0094**	**0.0000**	**0.0000**	**0.0021**	**0.0017**
	BRITS	0.0301	0.0385	0.0253	**0.0000**	**0.0000**	0.0046	0.0036
	CSDI	0.0163	0.0226	0.0175	**0.0000**	**0.0000**	0.0027	0.0022
(0.005, 0.5, 0.99)	Deletion	–	–	0.2960	0.0010	0.0005	0.0687	0.0268
	Zero	0.0321	0.0463	0.2238	0.0005	0.0003	0.0483	0.0172
	KNN	0.0161	0.0261	0.0991	0.0001	**0.0001**	0.0168	0.0095
	LowRank	0.0129	0.0200	**0.0279**	**0.0000**	**0.0001**	0.0078	0.0067
	EM	**0.0118**	**0.0189**	0.0319	**0.0000**	**0.0001**	**0.0077**	**0.0065**
	BRITS	0.0122	0.0197	0.0401	**0.0000**	**0.0001**	0.0082	0.0069
	CSDI	0.0136	0.0233	0.0751	**0.0000**	**0.0001**	0.0132	0.0087
(0.04, 0.2, 0.95)	Deletion	–	–	0.5325	0.0043	0.0018	0.1079	0.0491
	Zero	0.0143	0.0219	0.2254	0.0010	0.0007	0.0375	0.0143
	KNN	0.0092	0.0141	0.0562	0.0003	0.0002	0.0092	0.0067
	LowRank	0.0083	0.0125	0.0253	**0.0002**	**0.0001**	0.0068	0.0052
	EM	**0.0077**	**0.0117**	**0.0217**	**0.0002**	**0.0001**	**0.0062**	**0.0048**
	BRITS	0.0078	0.0118	0.0231	**0.0002**	**0.0001**	0.0064	0.0049
	CSDI	0.0078	0.0122	0.0374	**0.0002**	**0.0001**	0.0078	0.0061
(0.05, 0.5, 0.9)	Deletion	–	–	0.6318	0.0056	0.0023	0.1230	0.0630
	Zero	0.0174	0.0249	0.5125	0.0030	0.0017	0.0843	0.0244
	KNN	0.0103	0.0161	0.1602	0.0010	0.0004	0.0258	0.0124
	LowRank	0.0091	0.0139	**0.0483**	**0.0004**	**0.0002**	0.0110	0.0091
	EM	**0.0084**	**0.0130**	0.0554	0.0008	0.0003	**0.0107**	**0.0087**
	BRITS	0.0106	0.0162	0.2083	0.0015	0.0006	0.0256	0.0126
	CSDI	0.0086	0.0137	0.0542	0.0008	0.0003	0.0165	0.0109
(0.2, 1, 0.9)	Deletion	–	–	–	–	–	–	–
	Zero	0.0149	0.0219	0.7677	0.0070	0.0042	0.1201	0.0333
	KNN	–	–	–	–	–	–	–
	LowRank	0.0137	0.0212	0.6525	**0.0062**	0.0029	0.1111	0.0316
	EM	**0.0134**	**0.0210**	0.6634	0.0065	0.0031	0.1112	0.0315
	BRITS	0.0136	0.0214	**0.6480**	0.0063	**0.0025**	0.1069	0.0299
	CSDI	**0.0134**	**0.0210**	0.6559	0.0064	0.0028	**0.1065**	**0.0298**

3.1 Downstream evaluation: the optimal portfolio

We now turn to the downstream evaluation of Section 2.3.1. For every imputed panel, and for the uncorrupted one, we estimate the tangency portfolio and realise it on the true returns. Tables 3.4, 3.5 and 3.6 report its annualised mean, variance and Sharpe ratio, together with the gross distance $\|\Delta w\|_1$ between the estimated weights and the oracle's. The oracle is the same in every row, with an annualised mean of 0.361, a variance of 0.049 and a Sharpe ratio of 1.636. Since it is the best in-sample allocation, no imputation can beat its Sharpe, and the only question is how close each one comes. The two naive baselines are kept out of the tables and discussed at the end, so that the comparison stays among the imputation methods themselves.

Table 3.4: Optimal-portfolio evaluation under sparse missingness. Realised annualised mean, variance and Sharpe ratio of the tangency portfolio estimated from each panel, together with $\|\hat{w} - \hat{w}_{\text{True}}\|_1$, and the oracle estimated on the uncorrupted data.

Level	Model	Mean	Variance	Sharpe	$\|\Delta w\|_1$
0.001	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.362	0.0491	1.635	0.30
	LowRank	0.362	0.0490	1.635	0.29
	EM	0.362	0.0491	1.634	0.30
	BRITS	0.362	0.0492	1.634	0.33
	CSDI	0.362	0.0489	**1.635**	0.30
0.01	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.358	0.0486	1.623	1.11
	LowRank	0.360	0.0492	1.624	1.08
	EM	0.361	0.0493	1.625	1.03
	BRITS	0.359	0.0487	1.625	0.94
	CSDI	0.361	0.0492	**1.626**	1.02
0.05	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.373	0.0564	1.569	2.92
	LowRank	0.391	0.0625	1.565	3.01
	EM	0.391	0.0623	1.567	3.06
	BRITS	0.384	0.0597	1.572	2.81
	CSDI	0.396	0.0627	**1.580**	2.89
0.1	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.351	0.0554	1.490	4.07
	LowRank	0.375	0.0603	1.529	3.33
	EM	0.377	0.0607	1.531	3.40
	BRITS	0.371	0.0577	**1.542**	3.10
	CSDI	0.373	0.0597	1.527	3.03
0.3	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.422	0.1086	**1.279**	8.43
	LowRank	0.556	0.1970	1.253	12.21
	EM	0.581	0.2130	1.259	13.76
	BRITS	0.528	0.1852	1.226	12.61
	CSDI	0.531	0.1723	1.279	11.33

At the lightest levels there is nothing to separate them. Under sparse missingness at

Table 3.5: Optimal-portfolio evaluation under contiguous-gap missingness. Realised annualised mean, variance and Sharpe ratio of the tangency portfolio estimated from each panel, together with $\|\hat{w} - \hat{w}_{\text{True}}\|_1$, and the oracle estimated on the uncorrupted data.

Level	Model	Mean	Variance	Sharpe	$\|\Delta w\|_1$
(20, 10)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.363	0.0492	1.635	0.16
	LowRank	0.362	0.0491	1.635	0.11
	EM	0.362	0.0491	1.635	0.10
	BRITS	0.362	0.0490	**1.635**	0.10
	CSDI	0.362	0.0489	1.635	0.11
(200, 10)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.363	0.0497	1.627	0.87
	LowRank	0.365	0.0503	1.627	0.84
	EM	0.363	0.0499	1.628	0.84
	BRITS	0.362	0.0494	**1.629**	0.73
	CSDI	0.364	0.0499	1.628	0.79
(1000, 10)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.372	0.0557	1.575	2.24
	LowRank	0.369	0.0537	1.593	1.91
	EM	0.371	0.0539	**1.596**	1.77
	BRITS	0.369	0.0540	1.588	1.99
	CSDI	0.374	0.0555	1.589	1.97
(1000, 20)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.380	0.0644	1.500	4.31
	LowRank	0.386	0.0612	**1.560**	3.14
	EM	0.379	0.0598	1.549	3.40
	BRITS	0.358	0.0529	1.555	3.20
	CSDI	0.381	0.0611	1.542	3.52
(2000, 30)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.329	0.0767	1.186	6.52
	LowRank	0.465	0.1439	1.225	9.14
	EM	0.478	0.1460	1.250	10.17
	BRITS	0.423	0.1098	**1.276**	7.50
	CSDI	0.643	0.2937	1.186	14.48

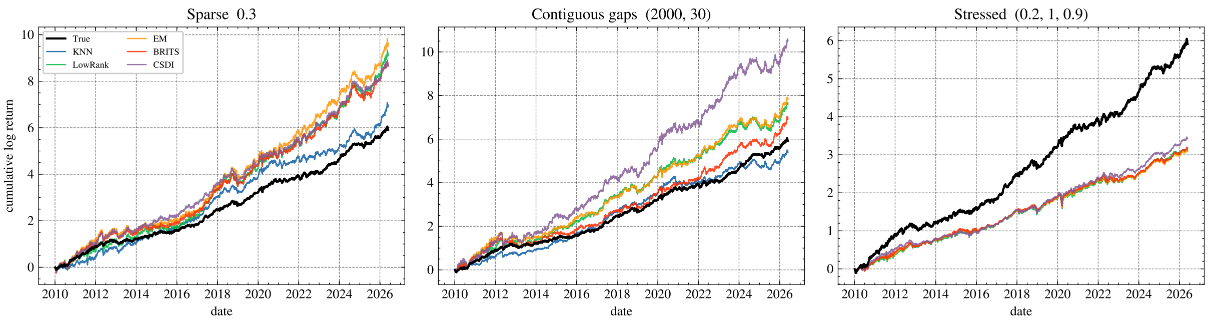

Figure 3.1: Realised cumulative log return of the tangency portfolio at the heaviest level of each missingness pattern. The oracle, estimated on the uncorrupted panel, is drawn in black; the five imputers are estimated on the corrupted panels and realised on the same true returns. Under the two MCAR patterns (left, centre) the imputed portfolios drift above the oracle by taking on more risk, while under the MNAR stressed pattern (right) they fall short of it. The two baselines, off the vertical scale, are omitted.

Table 3.6: Optimal-portfolio evaluation under stressed missingness. Realised annualised mean, variance and Sharpe ratio of the tangency portfolio estimated from each panel, together with $\|\hat{w} - \hat{w}_{\text{True}}\|_1$, and the oracle estimated on the uncorrupted data.

Level	Model	Mean	Variance	Sharpe	$\|\Delta w\|_1$
(0, 0.1, 0.99)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.347	0.0459	1.619	1.13
	LowRank	0.356	0.0477	1.629	0.76
	EM	0.356	0.0478	**1.631**	0.60
	BRITS	0.345	0.0452	1.620	1.20
	CSDI	0.360	0.0487	1.630	0.66
(0.005, 0.5, 0.99)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.328	0.0439	1.565	2.50
	LowRank	0.353	0.0483	1.606	1.58
	EM	0.353	0.0480	**1.612**	1.44
	BRITS	0.349	0.0476	1.602	1.78
	CSDI	0.375	0.0572	1.566	2.91
(0.04, 0.2, 0.95)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.359	0.0543	1.542	3.16
	LowRank	0.365	0.0529	1.586	2.09
	EM	0.370	0.0545	1.587	2.00
	BRITS	0.369	0.0536	**1.593**	1.91
	CSDI	0.392	0.0615	1.580	2.62
(0.05, 0.5, 0.9)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.346	0.0597	1.418	4.64
	LowRank	0.359	0.0576	1.494	3.50
	EM	0.359	0.0568	1.506	3.45
	BRITS	0.243	0.0347	1.304	4.49
	CSDI	0.354	0.0551	**1.508**	3.50
(0.2, 1, 0.9)	True (oracle)	0.361	0.0488	1.636	–
	KNN	–	–	–	–
	LowRank	0.191	0.0255	1.195	5.30
	EM	0.189	0.0245	1.207	5.22
	BRITS	0.193	0.0255	1.212	5.43
	CSDI	0.209	0.0266	**1.281**	5.22

rate 0.001, and under the shortest gaps, every imputer reproduces the oracle Sharpe to within a thousandth and moves the weights by less than 0.35 in gross terms. The gap opens only as the missingness grows, and even then the five methods stay close to one another, with no single model leading throughout: CSDI is marginally ahead on the light sparse levels, EM and low-rank completion on the gaps, EM again on most of the stressed levels, and the best entries at the easy levels are little more than ties. What separates the methods is less who leads than who fails. EM and low-rank completion are the steadiest, never the worst at any level. BRITS keeps pace until the stressed level (0.05, 0.5, 0.9), where it turns too cautious and its Sharpe falls to 1.30, against the 1.51 of EM. KNN holds the tightest weights under heavy sparse missingness but is left without donors at the most extreme stressed level and cannot be computed there. CSDI is the closest of all at that same extreme level, and yet at the heaviest gap it takes on the most risk of any method, with a realised variance of 0.29 and a weight distance above 14.

That last case is an instance of the general pattern in how the portfolios degrade. As the missingness grows the imputed panels do not lose a little Sharpe gracefully; their realised variance climbs well above the oracle's, from 0.049 to values past 0.2 at the heaviest sparse and gap levels, and the weights swell to many times their original size. This is the error maximisation that mean-variance optimisation is known for. A covariance matrix that the imputation has distorted, once inverted, produces an over-confident portfolio that takes large offsetting positions the true second moments never supported. The realised mean rises together with the variance, so in a market that trended upward over the sample these aggressive portfolios finish higher in level than the oracle while carrying far more risk, which the Sharpe ratio correctly discounts.

Figure 3.1 makes the two regimes visible at the heaviest level of each pattern. Under the two MCAR patterns the imputed portfolios end above the oracle, with visibly larger swings along the way: the distorted covariance inflates risk, and in this sample the extra risk happened to pay. The stressed pattern is the mirror image. There the missing dates are the turbulent ones, the imputers never see the volatility they would need to price, and their portfolios fall short of the oracle outright, ending at little more than half of its cumulative return. CSDI comes closest, but the shortfall that none of them close is the one the covariance error already exposed in Section 2.3: under MNAR the information needed to rebuild the portfolio is missing from the observed panel, and no imputer can supply it.

The baselines show, from below, how much the imputation matters once a model is built on top of it. Mean replacement trails the conditional methods at almost every level, the exception being the very heaviest, where its heavy damping of the covariance, a liability everywhere else, accidentally restrains the over-leverage that afflicts the others. Complete-case deletion is worse, and in a way the reconstruction metrics could only suggest. Estimating the covariance from the surviving dates alone leaves it so far from the truth that the optimiser levers into unstable portfolios: the realised variance is already more than three times the oracle's at sparse rate 0.05, the Sharpe falls to almost zero at rate 0.1, and at the gap level (1000, 20) the portfolio's mean return turns negative. At the heaviest levels deletion cannot be evaluated at all, too few complete dates surviving to estimate a covariance. A method that reads as merely inefficient in the metric tables can, once it feeds a portfolio, produce an allocation that is dangerously unstable.

3.2 Downstream evaluation: the factor structure

The second downstream model is the one of Section 2.3.1. From every imputed panel, and from the uncorrupted one, we extract the principal components of the estimated covariance and compare the resulting factor model with the true one. Tables 3.7, 3.8 and 3.9 report the share of the panel variance carried by the first component and by the first five, together with the subspace distances D_1 and D_5. On the uncorrupted panel the first principal component carries 41.74% of the variance and the first five carry 60.02%, and these are the values every row is read against. As with the portfolio, the two naive baselines are kept out of the tables and discussed at the end, so that the comparison stays among the imputation methods themselves.

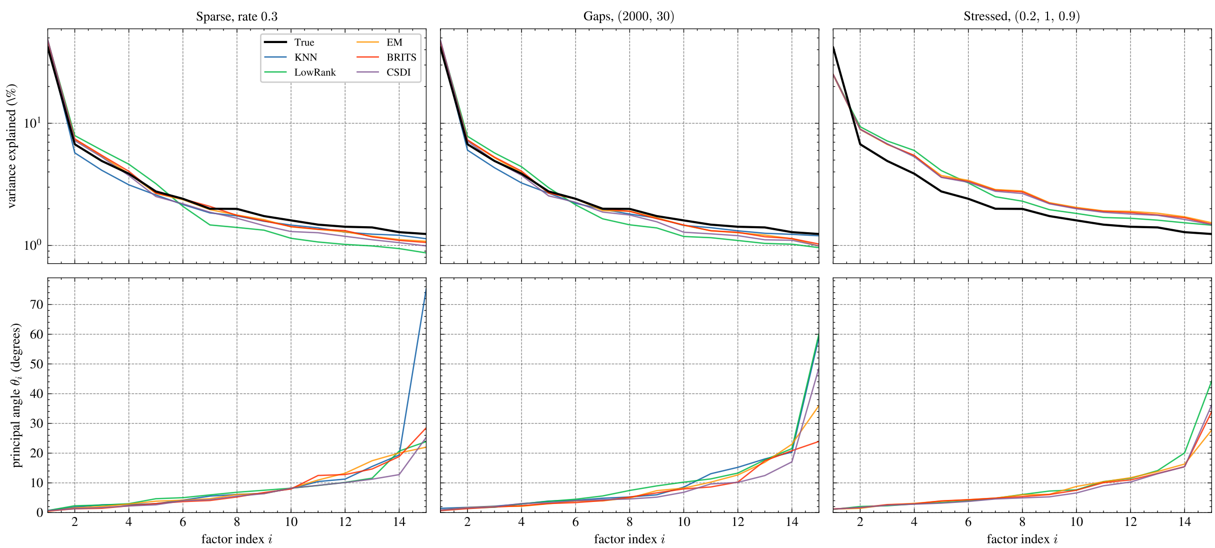

Figure 3.2: Factor structure at the heaviest level of each missingness pattern. Top: share of the panel variance carried by each of the first fifteen principal components, on a logarithmic scale, with the uncorrupted panel in black. Bottom: the fifteen principal angles between the true and the estimated fifteen-factor spaces, in increasing order. The leading directions are recovered to within a few degrees by every method, and the rotation grows with the depth of the factor.

The first factor survives almost everything. Under sparse missingness at rate 0.3, where a third of the panel is gone, D_1 stays below 0.026 for every method, an angle of less than a degree and a half between the estimated market factor and the true one. The five-factor

Table 3.7: Factor-structure evaluation under sparse missingness. Share of the panel variance carried by the first principal component and by the first five, and normalised subspace distance between the estimated and the true leading eigenspaces at one and five factors.

Level	Model	λ_1 (%)	Top-5 (%)	D_1	D_5
0.001	True (reference)	41.74	60.02	–	–
	KNN	41.76	60.04	0.0006	0.0051
	LowRank	41.77	60.06	0.0006	**0.0036**
	EM	41.77	60.06	0.0005	0.0040
	BRITS	41.77	60.05	0.0005	0.0046
	CSDI	41.77	60.06	0.0006	0.0041
0.01	True (reference)	41.74	60.02	–	–
	KNN	41.87	60.09	0.0030	0.0143
	LowRank	41.91	60.26	0.0026	0.0195
	EM	41.91	60.22	0.0026	0.0126
	BRITS	41.87	60.16	0.0034	0.0154
	CSDI	41.91	60.22	0.0027	**0.0126**
0.05	True (reference)	41.74	60.02	–	–
	KNN	42.28	60.32	0.0082	0.0364
	LowRank	42.64	61.37	0.0068	0.0809
	EM	42.64	61.19	0.0062	0.0440
	BRITS	42.59	61.14	0.0054	0.0347
	CSDI	42.72	61.18	0.0062	**0.0323**
0.1	True (reference)	41.74	60.02	–	–
	KNN	43.00	60.35	0.0133	**0.0436**
	LowRank	43.60	62.56	0.0100	0.2551
	EM	43.59	62.05	0.0094	0.1120
	BRITS	43.47	61.96	0.0100	0.1034
	CSDI	43.73	62.03	0.0122	0.1014
0.3	True (reference)	41.74	60.02	–	–
	KNN	45.62	61.19	0.0252	**0.1434**
	LowRank	47.56	69.35	0.0215	0.3914
	EM	47.88	67.16	0.0152	0.2039
	BRITS	47.09	66.74	0.0161	0.2376
	CSDI	48.54	67.28	0.0211	0.1954

Table 3.8: Factor-structure evaluation under contiguous-gap missingness. Share of the panel variance carried by the first principal component and by the first five, and normalised subspace distance between the estimated and the true leading eigenspaces at one and five factors.

Level	Model	λ_1 (%)	Top-5 (%)	D_1	D_5
(20, 10)	True (reference)	41.74	60.02	–	–
	KNN	41.75	60.02	0.0008	0.0040
	LowRank	41.75	60.03	0.0004	0.0033
	EM	41.75	60.03	0.0005	0.0034
	BRITS	41.75	60.03	0.0004	0.0029
	CSDI	41.75	60.03	0.0003	**0.0027**
(200, 10)	True (reference)	41.74	60.02	–	–
	KNN	41.90	60.15	0.0034	0.0117
	LowRank	41.93	60.26	0.0031	0.0207
	EM	41.92	60.24	0.0021	0.0117
	BRITS	41.92	60.24	0.0024	0.0115
	CSDI	41.92	60.24	0.0028	**0.0100**
(1000, 10)	True (reference)	41.74	60.02	–	–
	KNN	42.29	60.21	0.0089	0.0376
	LowRank	42.42	61.07	0.0053	0.0334
	EM	42.42	60.88	0.0047	**0.0249**
	BRITS	42.37	60.84	0.0046	0.0270
	CSDI	42.43	60.85	0.0040	0.0291
(1000, 20)	True (reference)	41.74	60.02	–	–
	KNN	42.93	60.62	0.0179	0.0525
	LowRank	43.63	62.50	0.0143	0.1892
	EM	43.52	62.14	0.0097	0.0550
	BRITS	43.44	62.08	0.0104	0.0508
	CSDI	43.78	62.22	0.0158	**0.0453**
(2000, 30)	True (reference)	41.74	60.02	–	–
	KNN	44.63	60.92	0.0511	0.1365
	LowRank	46.84	67.76	0.0375	0.3291
	EM	46.95	66.09	0.0262	0.1273
	BRITS	46.36	65.70	0.0252	**0.1177**
	CSDI	47.94	66.31	0.0465	0.1435

Table 3.9: Factor-structure evaluation under stressed missingness. Share of the panel variance carried by the first principal component and by the first five, and normalised subspace distance between the estimated and the true leading eigenspaces at one and five factors.

Level	Model	λ_1 (%)	Top-5 (%)	D_1	D_5
(0, 0.1, 0.99)	True (reference)	41.74	60.02	–	–
	KNN	41.31	59.70	0.0106	0.0276
	LowRank	41.75	60.12	0.0079	0.0152
	EM	41.73	60.07	0.0063	**0.0145**
	BRITS	41.45	59.73	0.0176	0.0317
	CSDI	42.06	60.23	0.0088	0.0160
(0.005, 0.5, 0.99)	True (reference)	41.74	60.02	–	–
	KNN	39.81	58.28	0.0309	0.1055
	LowRank	41.92	60.67	0.0180	**0.0555**
	EM	41.80	60.14	0.0178	0.0950
	BRITS	41.56	59.92	0.0203	0.0900
	CSDI	43.62	61.24	0.0316	0.1166
(0.04, 0.2, 0.95)	True (reference)	41.74	60.02	–	–
	KNN	41.68	59.49	0.0169	0.0680
	LowRank	42.73	61.36	0.0149	0.0714
	EM	42.66	61.03	0.0126	**0.0327**
	BRITS	42.58	60.96	0.0132	0.0368
	CSDI	43.27	61.34	0.0170	0.0383
(0.05, 0.5, 0.9)	True (reference)	41.74	60.02	–	–
	KNN	40.27	57.71	0.0398	0.1131
	LowRank	43.50	63.34	0.0268	0.3293
	EM	43.29	61.77	0.0232	**0.0843**
	BRITS	40.36	59.06	0.1407	0.1219
	CSDI	44.63	62.49	0.0303	0.0964
(0.2, 1, 0.9)	True (reference)	41.74	60.02	–	–
	KNN	–	–	–	–
	LowRank	25.12	51.75	0.1186	0.3814
	EM	24.81	49.58	0.1148	0.1343
	BRITS	25.19	50.00	0.1106	**0.1184**
	CSDI	25.01	49.66	0.1141	0.1340

space is a different matter: at the same level D_5 runs from 0.14 to 0.39, between six and eighteen times larger. The damage is not spread evenly over the leading factors but concentrated at the point where the spectrum stops separating them. The fifth and sixth true eigenvalues carry 2.77% and 2.41% of the variance, close enough that little in the data distinguishes their directions, and it is exactly there that the imputations diverge. At sparse rate 0.3 the first four directions are recovered to within about six degrees by KNN, EM, BRITS and CSDI, while their fifth is off by between 16.5 and 31.4 degrees. The angle profiles in Figure 3.2 show the same thing over a longer range: the rotation stays small through the first eight or ten factors and then climbs steeply.

The spectrum is distorted in a consistent direction. Every conditional imputer raises the weight of the first factor, from 41.74% to between 45.6% and 48.5% at sparse rate 0.3, and raises the five-factor share from 60.02% to between 61.2% and 69.3%. The reason is visible in what these methods do. An imputed value is a conditional mean, a combination of the assets observed on the same date, so it lies closer to the dominant directions of the panel than a real return does, and the more entries are filled the more the panel comes to look like its own factor model. The top row of Figure 3.2 shows the effect across the whole spectrum: under the two MCAR patterns the imputed curves sit above the true one at the head and below it in the tail. A downstream user reading these panels would conclude that the market is more tightly coupled than it is, and would underestimate the idiosyncratic variance that a hedge has to absorb.

Among the methods, low-rank completion behaves in a way worth isolating. It has the largest D_5 at ten of the fifteen levels, and at eight of the nine heaviest, reaching 0.39 under sparse missingness at rate 0.3 and 0.33 at the stressed level (0.05, 0.5, 0.9), against 0.21 and 0.08 for EM at the same levels. Its five-factor variance share is also the most inflated, 69.3% against a true 60.0%. The direction of this result is worth stating plainly, because it is the opposite of what the method's construction would suggest. Low rank completion is the one imputer that explicitly imposes a factor structure, and an evaluation of factor structure might be expected to favour it. What happens instead is that truncating the spectrum removes precisely the weakly separated middle directions: at sparse rate 0.3 its fifth factor is rotated by 55.8 degrees, where the worst of the other methods is at 31.4. It recovers the market factor as well as anyone and pays for it deeper down.

BRITS fails once, and the failure is informative. At the stressed level (0.05, 0.5, 0.9) its D_1 is 0.141, an angle of eight degrees on the leading factor, where the other four methods stay between 0.023 and 0.040, under two and a half degrees. This is the same level at which its portfolio Sharpe collapsed to 1.30 in Table 3.6 while EM and CSDI held above 1.50. The portfolio evaluation registered that something had gone wrong but could not say what; the factor evaluation locates it, in a rotation of the market factor itself, which no amount of care further down the spectrum can compensate.

At the most extreme stressed level the ranking dissolves and all the methods fail together. The first factor falls from 41.74% to about 25% and the five-factor share from 60.02% to about 50%, for low-rank completion, EM, BRITS and CSDI alike, with D_1 between 0.111 and 0.119 for all of them and KNN left without donors. The mechanism explains the uniformity. Stressed missingness removes the most turbulent dates, and those are the dates on which correlations rise and the market factor dominates, so what has been deleted is not a random part of the panel but the part that carries the factor structure. The right-hand column of Figure 3.2 shows every imputed spectrum lying flatter than the truth. This is the same conclusion the portfolio reached at this level, and the same one the covariance error of Section 2.3 reached before it: under MNAR the information is not in the observed panel, so no method can recover it and none of them separate.

The baselines fail in opposite directions, which is itself instructive. Mean replacement dilutes the factor structure rather than concentrating it, taking the first component down to 30.6% at sparse rate 0.3 and to 18.1% at the extreme stressed level, since a constant inserted in place of a return is uncorrelated with everything else in the cross section. Complete-case deletion is erratic in a way the reconstruction metrics never suggested. At the gap level (1000, 20) only 33 dates survive and its leading eigenvector is nearly orthogonal to the true one, $D_1 = 0.977$; at (2000, 30) three dates survive and five components exhaust the variance of the panel entirely; at sparse rate 0.3 and at the extreme stressed level no complete date survives at all and nothing can be estimated. Between them the two baselines bracket the failure: one flattens the factor structure, the other invents one.

The two downstream evaluations meet here. Since $\hat{\boldsymbol{\Sigma}}^{-1} = \sum_i \hat{\lambda}_i^{-1} \hat{v}_i \hat{v}_i^\top$, the inverse the portfolio depends on weights each direction by the reciprocal of its variance, so it loads most heavily on the trailing eigenvectors, which are exactly the directions the angle profiles show to be least reliable. The error maximisation of Section 3.1 is therefore not a separate defect of mean-variance optimisation. It is the distortion measured here, seen through an inverse that magnifies whatever the imputation has left in the parts of the spectrum it understands worst.

3.3 Computational costs

The scores above say nothing about what each model costs to produce, so we close the chapter with the computational side of the experiments. Everything in this thesis, training included, was run on a single consumer machine, an Apple Mac mini with the M4 chip (four performance and six efficiency cores) and 16 GB of unified memory. The neural models run on the CPU build of PyTorch, so no GPU acceleration is involved. Both networks are deliberately small, with roughly 1.2×10^5 trainable parameters for BRITS and 4.7×10^4 for CSDI, which means that the costs reported here come from the algorithms themselves and not from the size of the models.

The classical methods have no training stage and a negligible running time: the whole family fills a 4125×50 panel in a few seconds, and EM, the most expensive of the group with its fifty covariance sweeps, needs about one second. The neural models are on another scale. Training on a single corrupted panel took just under an hour for BRITS (10^4 epochs) and just under four hours for CSDI (2 000 epochs), so covering the fifteen levels of Section 2.1.1 required roughly seventy hours of computation, four fifths of which were spent on CSDI. CSDI is by far the most expensive model at imputation time as well. BRITS fills the panel with a single forward pass in a fraction of a second, while a CSDI imputation is the median of 32 stochastic draws, each of which runs the full 50-step reverse diffusion over every window of the panel. This takes roughly twenty-four minutes per panel, and the evaluation sweep behind Tables 3.1, 3.2 and 3.3 ran for about six hours, almost all of them spent sampling from CSDI. In view of the tables above, this cost buys very little: EM reaches the same quality for about one second per panel, four orders of magnitude less than CSDI, which rarely beats it.

Chapter 4

Conclusion

4.1 Summary

This thesis measured what is lost when the missing entries of a panel of daily equity returns are filled in. Fifty S&P 500 constituents over sixteen years were corrupted under three mechanisms at five severities each, reconstructed by seven methods, and scored both against the truth directly and through two models a user would build on the result. We return to the four questions of Section 1.3.

The mechanism matters more than the amount. Under the two random patterns the methods hold up even when a third of the panel is gone: at sparse rate 0.3 the market factor is still recovered to within a degree and a half of the true one. Under the stressed pattern, which removes turbulent dates, the same methods break down at far smaller shares of missing data. At the heaviest level none brings the relative covariance error below 0.6, the first principal component falls from 41.74% to about 25%, and the realised portfolios end at little more than half of the oracle's cumulative return, with the five imputers indistinguishable from one another. What has been deleted there is not a random part of the sample but the part that carries the covariance structure, and no model can recover from the observed panel what the observed panel does not contain. A comparable share of missing cells is not a comparable problem.

The deep models do not earn their cost. EM is rarely beaten in the metric tables and never by a wide margin, CSDI matches it, and BRITS falls behind at two of the stressed levels. EM fills a panel in about a second, while CSDI needs some twenty-four minutes of sampling after four hours of training, four orders of magnitude more for a reconstruction that is not better. The comparison is tied to the training budget and the hardware of Section 3.3, but the size of the gap leaves a good deal of room before the conclusion would turn.

Pointwise accuracy is a poor proxy for the quality of an imputed panel. Under the MCAR patterns the MAE of mean replacement is only about fifty percent worse than the best, while its covariance error is close to an order of magnitude worse than EM's, so a method can look acceptable entry by entry and still distort the geometry of the panel. The distortion is also systematic rather than noisy. Every conditional imputer raises the share of variance carried by the first factor, from 41.74% to between 45.6% and 48.5% at sparse rate 0.3, because the value it inserts is a conditional mean and therefore lies closer to the dominant directions of the panel than a real return does. The more entries are filled, the more the panel comes to look like its own factor model, and no pointwise metric registers this.

The rankings largely survive downstream, and the exceptions carry the most information. Low-rank completion is among the most accurate methods entry by entry and has the largest five-factor subspace distance at ten of the fifteen levels, reaching 0.39 at sparse rate 0.3 against 0.21 for EM, since truncating the spectrum discards exactly the weakly separated middle directions. Complete-case deletion reads as merely inefficient in the metric tables and produces allocations whose Sharpe ratio falls to almost zero at sparse rate 0.1 and whose mean return turns negative at the gap level (1000, 20). The two downstream evaluations are themselves connected: since $\hat{\boldsymbol{\Sigma}}^{-1} = \sum_i \hat{\lambda}_i^{-1} \hat{v}_i \hat{v}_i^{\top}$, the portfolio weights each direction by the reciprocal of its variance and so loads most heavily on the trailing eigenvectors, which are the directions the imputations recover worst. The error maximisation seen in Section 3.1 is that distortion viewed through an inverse that magnifies it.

The practical reading is that an imputation should be scored against the use it is put to. Reconstruction error orders the methods correctly at the easy levels, where the ordering hardly matters, and is least informative exactly where the choice of method is consequential.

4.2 Limitations

The evidence comes from a single panel: fifty liquid large-cap equities from one market over one period, at daily frequency. Fifty assets against 4125 dates is a comfortable ratio for covariance estimation, and a cross section closer in size to the sample length would make every estimate less stable and would likely widen the differences reported here. The missingness is injected rather than observed, so its mechanism is known and stationary, whereas real gaps arise from halts, delistings and illiquidity and follow no law the experimenter can write down; the stressed pattern is a stylised form of MNAR and not a model of any particular cause. Each configuration was run once, so the tables carry no estimate of sampling variation around the numbers they report. The two downstream evaluations both read the covariance matrix, which makes them complementary but not independent tests. Finally, all seven methods are smoothers. BRITS reads the panel in both directions and CSDI conditions on every observed value, so the results describe reconstruction after the fact and not what would be available to a user filling a gap in real time, with only the past in hand.

4.3 Future Work

The last of those limitations is the natural continuation. Restricting each imputer to the information available up to the date being filled turns the problem from smoothing into a filtering one and makes the cost of that restriction measurable, which is what a user working in real time needs to know and what the benchmarks in the literature do not report. A second direction is to widen the downstream evaluations beyond the second moment, with a Value-at-Risk backtest, a hedging exercise or a trading rule, since a criterion that does not pass through the covariance would test parts of the reconstruction that neither of the present evaluations reaches. A third is to carry the uncertainty of the imputation forward instead of discarding it. CSDI produces a predictive distribution that we collapsed to a median, and the inflation of the factor structure documented above is precisely the artefact that averaging conditional means creates, so a multiple-imputation treatment in which the downstream decision sees the spread rather than a point estimate should correct it. Beyond these, the same design applies directly to larger cross sections, lower frequencies and other asset classes, and eventually to panels whose gaps are real rather than injected, where the mechanism has to be inferred rather than assumed.

Bibliography

A. A. Afifi and R. M. Elashoff. Missing observations in multivariate statistics I: Review of the literature. *Journal of the American Statistical Association*, 61(315):595–604, 1966. doi: 10.1080/01621459.1966.10480891.

F. E. Allan and John Wishart. A method of estimating the yield of a missing plot in field experimental work. *The Journal of Agricultural Science*, 20(3):399–406, 1930. doi: 10.1017/S0021859600006912.

Theodore W. Anderson. Maximum likelihood estimates for a multivariate normal distribution when some observations are missing. *Journal of the American Statistical Association*, 52(278):200–203, 1957. doi: 10.1080/01621459.1957.10501379.

Marta Bańbura and Michele Modugno. Maximum likelihood estimation of factor models on datasets with arbitrary pattern of missing data. *Journal of Applied Econometrics*, 29 (1):133–160, 2014. doi: 10.1002/jae.2306.

Maurice S. Bartlett. Some examples of statistical methods of research in agriculture and applied biology. *Supplement to the Journal of the Royal Statistical Society*, 4(2):137–170, 1937. doi: 10.2307/2983644.

Fischer Black. Studies of stock price volatility changes. In *Proceedings of the 1976 Meetings of the American Statistical Association, Business and Economic Statistics Section*, pages 177–181, 1976.

Svetlana Bryzgalova, Sven Lerner, Martin Lettau, and Markus Pelger. Missing financial data. *Review of Financial Studies*, 38(3):803–882, 2025. doi: 10.1093/rfs/hhae036.

S. F. Buck. A method of estimation of missing values in multivariate data suitable for use with an electronic computer. *Journal of the Royal Statistical Society: Series B*, 22(2): 302–306, 1960. doi: 10.1111/j.2517-6161.1960.tb00375.x.

Emmanuel J. Candès and Benjamin Recht. Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9(6):717–772, 2009. doi: 10.1007/s10208-009-9045-5.

Wei Cao, Dong Wang, Jian Li, Hao Zhou, Lei Li, and Yitan Li. BRITS: Bidirectional recurrent imputation for time series. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Saumitra Choudhury, Kaspar Schindler, and Rama Cont. Time-series imputation with Wasserstein interpolation for optimal look-ahead-bias and variance tradeoff. *arXiv preprint arXiv:2102.12736*, 2021.

Daniel Commenges and Hélène Jacqmin-Gadda. Missing at random: a stochastic process perspective. *Biometrika*, 109(1):81–95, 2022. doi: 10.1093/biomet/asab011.

Rama Cont. Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001. doi: 10.1080/713665670.

Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1):1–38, 1977.

Yi Dong, Fan Sun, Wenjie Du, and Yuxuan Liang. Deep learning for multivariate time series imputation: A survey. *arXiv preprint arXiv:2402.04059*, 2024.

Catherine Doz, Domenico Giannone, and Lucrezia Reichlin. A quasi–maximum likelihood approach for large, approximate dynamic factor models. *Review of Economics and Statistics*, 94(4):1014–1024, 2012. doi: 10.1162/REST_a_00225.

Wenjie Du, David Côté, and Yan Liu. SAITS: Self-attention-based imputation for time series. *Expert Systems with Applications*, 219:119619, 2023. doi: 10.1016/j.eswa.2023.119619.

James Durbin and Siem Jan Koopman. *Time Series Analysis by State Space Methods*. Oxford University Press, Oxford, 2 edition, 2012.

Vincent Fortuin, Dmitry Baranchuk, Gunnar Rätsch, and Stephan Mandt. GP-VAE: Deep probabilistic time series imputation. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, 2020.

Joachim Freyberger, Björn Höppner, Andreas Neuhierl, and Michael Weber. Missing data in asset pricing panels. *Review of Financial Studies*, 38, 2025. doi: 10.1093/rfs/hhae003.

Domenico Giannone, Lucrezia Reichlin, and David Small. Nowcasting: The real-time informational content of macroeconomic data. *Journal of Monetary Economics*, 55(4): 665–676, 2008. doi: 10.1016/j.jmoneco.2008.05.010.

Andrew C. Harvey. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press, Cambridge, 1989.

Andrew C. Harvey and Richard G. Pierse. Estimating missing observations in economic time series. *Journal of the American Statistical Association*, 79(385):125–131, 1984. doi: 10.1080/01621459.1984.10477074.

Michael Healy and Michael Westmacott. Missing values in experiments analysed on automatic computers. *Journal of the Royal Statistical Society: Series C*, 5(3):203–206, 1956. doi: 10.2307/2986612.

James J. Heckman. Sample selection bias as a specification error. *Econometrica*, 47(1): 153–161, 1979. doi: 10.2307/1912352.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.

James Honaker and Gary King. What to do about missing values in time-series cross-section data. *American Journal of Political Science*, 54(2):561–581, 2010. doi: 10.1111/j.1540-5907.2010.00447.x.

Richard H. Jones. Maximum likelihood fitting of ARMA models to time series with missing observations. *Technometrics*, 22(3):389–395, 1980. doi: 10.1080/00401706.1980.10486171.

Rudolf E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960. doi: 10.1115/1.3662552.

Paul Kofman and Ian G. Sharpe. Using multiple imputation in the analysis of incomplete observations in finance. *Journal of Financial Econometrics*, 1(2):216–249, 2003. doi: 10.1093/jjfinec/nbg013.

Robert Kohn and Craig F. Ansley. Estimation, prediction, and interpolation for ARIMA models with missing data. *Journal of the American Statistical Association*, 81(395): 751–761, 1986. doi: 10.1080/01621459.1986.10478332.

Alexander Kreiss, Enno Mammen, and Wolfgang Polonik. Missing data in signal processing and machine learning: Models, methods and modern approaches. *arXiv preprint arXiv:2506.01696*, 2024.

Roderick J. A. Little. Robust estimation of the mean and covariance matrix from data with missing values. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 37 (1):23–38, 1988. doi: 10.2307/2347491.

Roderick J. A. Little and Donald B. Rubin. *Statistical Analysis with Missing Data*. John Wiley & Sons, Hoboken, NJ, 3 edition, 2019. doi: 10.1002/9781119482260.

Chuanhai Liu and Donald B. Rubin. The ECME algorithm: A simple extension of EM and ECM with faster monotone convergence. *Biometrika*, 81(4):633–648, 1994. doi: 10.1093/biomet/81.4.633.

Chuanhai Liu and Donald B. Rubin. ML estimation of the t distribution using EM and its extensions, ECM and ECME. *Statistica Sinica*, 5(1):19–39, 1995.

Thomas A. Louis. Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 44(2):226–233, 1982.

Jing Ma, Ruocheng Guo, Chenwei Chen, Aidong Zhang, and Jundong Li. Causal view of time series imputation: Some identification results on missing mechanism. *arXiv preprint arXiv:2505.07180*, 2025.

Roberto S. Mariano and Yasutomo Murasawa. A new coincident index of business cycles based on monthly and quarterly series. *Journal of Applied Econometrics*, 18(4):427–443, 2003. doi: 10.1002/jae.695.

Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952. doi: 10.1111/j.1540-6261.1952.tb01525.x.

Rahul Mazumder, Trevor Hastie, and Robert Tibshirani. Spectral regularization algorithms for learning large incomplete matrices. *Journal of Machine Learning Research*, 11:2287–2322, 2010.

Geoffrey J. McLachlan and Thriyambakam Krishnan. *The EM Algorithm and Extensions*. John Wiley & Sons, Hoboken, NJ, 2 edition, 2008.

Xiao-Li Meng and Donald B. Rubin. Maximum likelihood estimation via the ECM algorithm: A general framework. *Biometrika*, 80(2):267–278, 1993. doi: 10.1093/biomet/80.2.267.

Geert Molenberghs and Michael G. Kenward. *Missing Data in Clinical Studies*. John Wiley & Sons, Chichester, 2007.

Terence Orchard and Max A. Woodbury. A missing information principle: Theory and applications. In *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Theory of Statistics*, pages 697–715, Berkeley, CA, 1972. University of California Press.

James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866, 1994. doi: 10.1080/01621459.1994.10476818.

Donald B. Rubin. Inference and missing data. *Biometrika*, 63(3):581–592, 1976. doi: 10.2307/2335739.

Donald B. Rubin. Multiple imputation for nonresponse in surveys. 1987.

Donald B. Rubin. Multiple imputation after 18+ years. *Journal of the American Statistical Association*, 91(434):473–489, 1996. doi: 10.1080/01621459.1996.10476908.

Joseph L. Schafer. *Analysis of Incomplete Multivariate Data*. Chapman & Hall/CRC, London, 1997.

Myron Scholes and Joseph Williams. Estimating betas from nonsynchronous data. *Journal of Financial Economics*, 5(3):309–327, 1977. doi: 10.1016/0304-405X(77)90041-1.

Shaun R. Seaman and Ian R. White. Review of inverse probability weighting for dealing with missing data. *Statistical Methods in Medical Research*, 22(3):278–295, 2013. doi: 10.1177/0962280210395740.

Robert H. Shumway and David S. Stoffer. An approach to time series smoothing and forecasting using the EM algorithm. *Journal of Time Series Analysis*, 3(4):253–264, 1982. doi: 10.1111/j.1467-9892.1982.tb00349.x.

Robert F. Stambaugh. Analyzing investments whose histories differ in length. *Journal of Financial Economics*, 45(3):285–331, 1997. doi: 10.1016/S0304-405X(97)00020-2.

Daniel J. Stekhoven and Peter Bühlmann. Missforest—non-parametric missing value imputation for mixed-type data. *Bioinformatics*, 28(1):112–118, 2012. doi: 10.1093/bioinformatics/btr597.

James H. Stock and Mark W. Watson. Macroeconomic forecasting using diffusion indexes. *Journal of Business & Economic Statistics*, 20(2):147–162, 2002. doi: 10.1198/073500102317351921.

Yusuke Tashiro, Jiaming Song, Yang Song, and Stefano Ermon. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. In *Advances in Neural Information Processing Systems*, volume 34, pages 24804–24816, 2021.

Olga Troyanskaya, Michael Cantor, Gavin Sherlock, Pat Brown, Trevor Hastie, Robert Tibshirani, David Botstein, and Russ B. Altman. Missing value estimation methods for DNA microarrays. *Bioinformatics*, 17(6):520–525, 2001. doi: 10.1093/bioinformatics/17.6.520.

Stef van Buuren. *Flexible Imputation of Missing Data*. CRC Press, Boca Raton, FL, 2 edition, 2018. URL `https://stefvanbuuren.name/fimd/`.

Stef van Buuren and Karin Groothuis-Oudshoorn. mice: Multivariate imputation by chained equations in R. *Journal of Statistical Software*, 45(3):1–67, 2011. doi: 10.18637/jss.v045.i03.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Samuel S. Wilks. Moments and distributions of estimates of population parameters from fragmentary samples. *The Annals of Mathematical Statistics*, 3(3):163–195, 1932. doi: 10.1214/aoms/1177732885.

C. F. Jeff Wu. On the convergence properties of the EM algorithm. *The Annals of Statistics*, 11(1):95–103, 1983. doi: 10.1214/aos/1176346060.

Frank Yates. The analysis of replicated experiments when the field results are incomplete. *Empire Journal of Experimental Agriculture*, 1(2):129–142, 1933.

Jinsung Yoon, James Jordon, and Mihaela van der Schaar. GAIN: Missing data imputation using generative adversarial nets. In *Proceedings of the 35th International Conference on Machine Learning*, 2018.

Kyunghyun Cho Zhengping Che, Sanjay Purushotham and Yan Liu. Recurrent neural networks for multivariate time series with missing values. *arXiv preprint arXiv:1606.01865*, 2016.

List of Figures

List of Tables